

KARNATAKA 1ST PUC BOARD PREPARATION

Chapter 2: Biological Classification Official Blueprint Updated Study Guide

BAIO LEARNING SYSTEM | Narendra Naslai - Founder & CEO

Subject: Biology (36) | Weightage: 4 Marks

1. OFFICIAL KARNATAKA 1ST PUC CHAPTERWISE ALLOTMENT

PARAMETER	OFFICIAL ALLOTMENT DATA (CHAPTER 2)
Chapter Name	Chapter 2 - Biological Classification
Teaching Hours Allotted	5 Hours (Periods)
Official Marks Allotted	4 Marks
Page Numbers in Official Notes	Pages 7 – 15
Overall Paper Format	3 Hours Max Marks: 70 Total Allotted Weightage: 105 Marks (including choices)

Expected 4-Mark Question Distribution Configurations

- **Pattern A:** 1 Three-Mark Question (Part C) + 1 One-Mark MCQ/VSA (Part A). **Most Common**
- **Pattern B:** 1 Two-Mark Question (Part B) + 1 Two-Mark Question (Part B).
- **Pattern C:** 1 Two-Mark Question (Part B) + 2 One-Mark Questions (Part A MCQs/VSAs).

2. TOP HIGH-YIELD TOPICS TARGETED FOR 4 MARKS ALLOTMENT

TOPIC 1 [3-Mark Essential]: Comparative Analysis of Fungal Classes

Frequently asked directly as a 3-Mark question in Part C: "Give the salient differences between *Phycomycetes*, *Ascomycetes*, *Basidiomycetes*, and *Deuteromycetes*."

FEATURE	PHYCOMYCETES	ASCOMYCETES	BASIDIOMYCETES	DEUTEROMYCETES
Common Name	Algal Fungi	Sac-Fungi	Club / Bracket Fungi	Imperfect Fungi
Mycelium	Aseptate & Coenocytic	Branched & Septate	Branched & Septate	Branched & Septate
Asexual Spores	Zoospores / Aplanospores	Conidia (exogenous)	Generally absent	Conidia
Sexual Spores	Zygospores	Ascospores (endogenous)	Basidiospores (exogenous)	Absent (Unknown)
Examples	<i>Mucor</i> , <i>Rhizopus</i> , <i>Albugo</i>	<i>Penicillium</i> , <i>Aspergillus</i>	<i>Agaricus</i> , <i>Puccinia</i>	<i>Alternaria</i> , <i>Trichoderma</i>

TOPIC 2: Five Kingdom Classification & Criteria (2 or 3 Marks)

Core Question: "List the five kingdoms proposed by R.H. Whittaker and state the criteria used for this classification."

- **Five Kingdoms:** Monera, Protista, Fungi, Plantae, and Animalia.
- **Five Main Criteria:**
 1. Cell structure (Prokaryotic vs. Eukaryotic).

2. Body organisation (Unicellular vs. Multicellular/Tissue/Organ).
3. Mode of nutrition (Autotrophic vs. Heterotrophic).
4. Reproduction.
5. Phylogenetic relationships.

TOPIC 3: Viruses, Viroids, Prions & Lichens (2 or 3 Marks)

• Viruses vs. Viroids:

- **Viruses:** Contain genetic material (RNA or DNA) enclosed in a protein coat called a capsid.
- **Viroids:** Free infectious RNA lacking a protein coat, having a lower molecular weight (discovered by T.O. Diener).

• **Lichens:** Symbiotic association between an algal partner (**phycobiont**—autotrophic/photosynthetic) and a fungal partner (**mycobiont**—heterotrophic/provides shelter & nutrients). Excellent pollution indicators.

• **Prions:** Abnormally folded infectious proteins causing neurological diseases such as BSE (Mad Cow Disease) and CJD.

TOPIC 4: Archaeobacteria, Specialized Structures & Cyanobacteria (1 or 2 Marks)

• **Archaeobacteria Adaptations:** Specialized cell wall structure enables survival in extreme environments: *Halophiles* (salty areas), *Thermoacidophiles* (hot springs), and *Methanogens* (marshy areas / gut of ruminants).

• Specialized Structures:

- **Heterocysts:** Specialized, thick-walled cells in cyanobacteria (e.g., *Nostoc*, *Anabaena*) for atmospheric nitrogen fixation.
- **Diatomaceous Earth:** Cell wall deposits of diatoms embedded with silica; indestructible and used in polishing and oil filtration.

3. MODEL QUESTIONS MATCHED TO 4-MARK BLUEPRINT COMBINATION

Q1. Which organism is known as the chief producer in the oceans?

1 Mark (Part A MCQ)

Model Answer: Diatoms (Chrysophytes). [1 Mark]

Q2. Name the genetic material present in bacteriophages.

1 Mark (Part A VSA)

Model Answer: Double-stranded DNA (dsDNA). [1 Mark]

Q3. Differentiate between Phycobiont and Mycobiont in Lichens.

2 Marks (Part B)

Model Answer:

1. **Phycobiont:** The photosynthetic algal component that prepares food for the association. [1 Mark]
2. **Mycobiont:** The heterotrophic fungal component that provides shelter and absorbs mineral nutrients and water. [1 Mark]

Q4. Describe the three main steps involved in the sexual cycle of Fungi.

3 Marks (Part C)

Model Answer:

1. **Plasmogamy:** Fusion of protoplasts between two motile or non-motile gametes. [1 Mark]
2. **Karyogamy:** Fusion of two haploid nuclei to form a diploid nucleus. [1 Mark]
3. **Meiosis:** Reduction division in the zygote resulting in the formation of haploid spores. [1 Mark]

Q5. Mention three characteristic features of Kingdom Protista with examples.

3 Marks (Part C)

Model Answer:

1. **Cellular Nature:** Includes all single-celled eukaryotic organisms with a well-defined nucleus and membrane-bound organelles. [1 Mark]
2. **Habitat & Locomotion:** Primarily aquatic; move via flagella, cilia, or pseudopodia. [1 Mark]
3. **Diversity & Examples:** Shows diverse modes of nutrition (photosynthetic/heterotrophic); e.g., *Chrysophytes* (Diatoms), *Euglenoids* (*Euglena*), *Protozoans* (*Paramecium*). [1 Mark]

BLUEPRINT SCORING TIP FOR KARNATAKA 1ST PUC

Since Chapter 2 carries **4 Marks** with **5 Teaching Hours**, focus heavily on 3-mark conceptual questions from Kingdom Fungi (classes and sexual cycle), Five Kingdom criteria, or Virus/Viroid distinctions, alongside 1-mark objective questions on Cyanobacteria, Lichens, or Archaeobacteria examples.